CONSTRUCTION PLANNING & MANAGEMENT

Proposal for a Residential Complex

2000 Apartments — Type V(E) & Type VI(F)

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. Executive Summary

This proposal outlines the development of a residential complex delivering 2,000 apartments, split equally between Type V(E) and Type VI(F) units, designed in accordance with CPWD Plinth Area Rates (PAR) 2025 guidelines. The project integrates sustainable design practices, rigorous area and cost estimation, comprehensive amenity planning, structured resource scheduling, and milestone-based fund management to ensure efficient, on-time delivery.

2. Project Brief & Objectives

- Development of 12 G+20 towers housing up to 8 flats per floor.
- Inclusion of community amenities: club, health centre, banquet halls, and festival grounds.
- Multi-level parking accommodating 2,500 cars (residents and visitors).
- Sustainable design ensuring energy and water efficiency.

3. Assumptions & References

Apartment floor areas and servant units are based on CPWD Annexure 1 (2024 Column) plinth area values:

- Type V(E) main unit: 161.5 sqm; Servant unit: 25 sqm
- Type VI(F) main unit: 229 sqm; Servant unit: 25 sqm

Cost rates are applied as per CPWD Plinth Area Rates 2025:

- Residential RCC framed buildings: Rs. 24,410 per sqm
- Club and health centres: Rs. 31,990 per sqm
- Parking bays: Rs. 10,000 per car space
- Landscaping and external works as per CPWD rates

An additional 20% area is added on flats for circulation, common areas, lobbies, and services, based on CPWD guidelines and standard industry practice.

4. CPWD Data (Plinth Areas & Rates)

Residential RCC Framed Buildings

Base rate for 3.0 m floor height, up to 6 storeys: Rs. 24,410 per sqm.

Storey increment charges:

- 7th to 12th floor: additional Rs. 123 per sqm
- 13th to 18th floor: additional Rs. 246 per sqm
- 19th to 24th floor: additional Rs. 369 per sqm
- 25th to 30th floor: additional Rs. 492 per sqm

Clubs and Health Centres

Rate for non-residential buildings such as hospitals and clubhouses: Rs. 31,990 per sqm (reflective of hospital rates).

Multilevel Parking

Sensor-based car parking system: Rs. 10,000 per car bay.

Landscaping and External Works

- Landscaping operations: Rs. 578 per sqm
- Internal roads (bituminous topping): Rs. 1,940 per sqm

5. Apartment & Tower Planning

- 12 towers arranged to optimize solar orientation and ventilation.
- A maximum of 8 apartments per floor to maintain privacy and adequate corridor width.
- Towers separated by landscaped open spaces to ensure air circulation.
- Load-bearing cores for stairs and lifts integrated to maximize usable area.

Key Strengths of the Plan

- Comprehensive planning covering design, area estimation, cost, scheduling, and fund flow in adherence to CPWD PAR 2025 standards.
- Balanced apartment mix of Type V(E) and Type VI(F) units caters to diverse family sizes and preferences.
- Robust amenities — clubhouse, health centre, banquet halls, and festival grounds — promote community living and wellness.
- Optimized multi-level parking addresses resident and visitor needs while reducing urban congestion.
- Sustainability focus through solar power, rainwater harvesting, and energy-efficient systems.
- Phased construction with milestone-based fund flow supports timely execution and financial management.
- Realistic 30-month project timeline with parallel construction phases enhances feasibility.
- Compliance with regulatory norms for lifts, staircases, fire safety, and accessibility ensures occupant well-being.
- Use of updated CPWD rates ensures accurate budgeting and cost control.

6. Area Estimation

Apartment Areas

Flat Type	Main Unit Area (sqm)	Servant Unit Area (sqm)	Number of Units	Total Area (sqm)
Type V(E)	161.5	25	1,000	186,500
Type VI(F)	229	25	1,000	254,000
Subtotal				440,500
Additional Common Areas (20%)				88,100
Total Built-Up Area				528,600

Table 6.1 — Apartment area breakdown and total estimated built-up area.

Parking Area Calculation (MLCP — 3 Levels)

Parameter	Value	Unit / Notes
Number of Resident Cars	2,300	Based on 1.15 cars per flat
Number of Visitor Cars	200	As per problem statement
Total Required Parking	2,500	Car bays
Parking Space per Car	30	sqm, including circulation
Total Parking Area	75,000	sqm
Parking Levels	3	Multi-level parking floors
Footprint per Level	25,000	sqm
Cost per Car Bay	Rs. 10,000	CPWD PAR 2025 rate
Total Parking Cost Estimate	Rs. 2.5 crore	Rs. 25,000,000

Table 6.2 — Multi-level car parking sizing and cost basis.

Total resident car parking is derived by assuming 1.15 cars per apartment (2,000 × 1.15 = 2,300, rounded), to which 200 dedicated visitor spaces are added, giving a total requirement of 2,500 car bays. At a CPWD-recommended provision of 30 sqm per car — inclusive of circulation and ramps — the total parking area required is 75,000 sqm. Distributing this across 3 levels of a multi-level car park (MLCP) gives a footprint of 25,000 sqm per level, positioned near the residential towers.

7. Cost Estimation

Component	Quantity	Rate (Rs./unit)	Cost (Rs.)
Residential Building	528,600 sqm	24,410	12,901,026,000
Services & Amenities	(15% of base)	–	1,935,153,900
Club & Health Centre	5,000 sqm	31,990	159,950,000

Component	Quantity	Rate (Rs./unit)	Cost (Rs.)
Parking	2,500 bays	10,000	25,000,000
Landscaping	20,000 sqm	578	11,560,000
Internal Roads	20,000 sqm	1,940	38,800,000
TOTAL ESTIMATE			15,071,489,900

Table 7.1 — Consolidated project cost estimate (Rs. 1,507.15 crore).

Basis of Estimation

- Residential Building: 528,600 sqm of total built-up area at Rs. 24,410 per sqm (RCC framed residential rate).
- Services & Amenities: plumbing, electromechanical systems, internal electrification, and fire protection, taken at 15% of the base building cost.
- Club & Health Centre: approximately 5,000 sqm at Rs. 31,990 per sqm (non-residential / hospital-equivalent rate).
- Parking: 2,500 bays at Rs. 10,000 per bay for the sensor-based multi-level system.
- Landscaping: 20,000 sqm at Rs. 578 per sqm.
- Internal Roads: 20,000 sqm of bituminous topping at Rs. 1,940 per sqm.

8. Parking & External Works

- Parking configuration: multi-level car park (MLCP) across 3 levels, selected for its balance of cost and land-use efficiency.
- Dedicated visitor parking: 200 spaces.
- Supporting infrastructure: service roads, stormwater drains, sewerage, sewage treatment plant (STP), electrical room, and transformer area.

9. Amenities: Club, Health Centre & Others

Clubhouse

Swimming pool, library, yoga room, indoor games, a small theatre, banquet halls, and a restaurant.

Health Centre

Emergency ward with 10 beds and 4 OPDs for visiting physicians.

Community Spaces

Festival grounds, landscaped gardens (approximately 5,000 sqm), walkways, and gathering areas.

Parking

Multi-layered parking for 2,500 cars, including visitor bays.

Additional Facilities

Recreational, social, and wellness facilities designed to promote community living and well-being.

10. Project Schedule & Resource Plan

Phase	Duration (Months)	Notes
Mobilization	3	Site preparation, material procurement
Foundations	4	Includes deep foundation and piling
Superstructure	10	Main frame erection for towers
Interiors and Amenities	6	Electrical, plumbing, finishing
External Works	5	Roads, landscaping, parking completion
Testing & Commissioning	2	Quality checks, approvals, handover

Table 10.1 — Phase-wise project schedule totalling a 30-month timeline.

11. Fund Flow / Payment Milestones

Funds are released in stages aligned with project milestones, ensuring that payment correlates with work completed. An initial 10% advance covers mobilization, site setup, and early procurement. Following foundation completion, a further 10% is released to secure funding for critical base structural work. Up to 40% of funds are disbursed during superstructure work, reflecting its resource intensity, while interiors and finishing stages receive about 30% as work shifts to fit-outs, plumbing, electricals, and amenities. The final 10% covers external works, testing, inspections, and formal handover.

Stage / Milestone	% of Total Cost	Amount (Rs. Crores)	Timeline
Advance & Mobilization	10%	151	Month 0–3
Foundation Completion	10%	151	Month 4–7
Superstructure Completion	40%	603	Month 8–17
Interiors & Amenities	30%	452	Month 18–23
External Works & Handover	10%	151	Month 24–30

Table 11.1 — Milestone-based fund disbursement schedule.

Milestone-based payments manage cash flow effectively, reduce financial risk, and motivate timely completion. Regular, documented fund releases further enhance stakeholder confidence and project transparency.

12. Sustainability & Additional Measures

- Implementation of a 250 kWp solar energy system.
- Rainwater harvesting and reuse for landscaping.
- Green landscaping with native, drought-resistant plants.
- Use of energy-efficient systems and materials.
- Wastewater treatment and greywater recycling.

13. Conclusion

The proposed residential complex plan offers a fully integrated approach that balances functional design, cost-effectiveness, and sustainability with robust project management practices. Utilizing CPWD Plinth Area Rates 2025 has provided precise and up-to-date benchmarks for area calculation and financial estimation, ensuring realistic budgeting.

The detailed apartment and servant unit area breakdown, together with allowances for common areas, enables accurate space planning that meets regulatory and user requirements. Facility allocation — including the club, health centre, and multi-level parking — supports resident needs comprehensively while optimizing site utilization.

A milestone-based fund flow aligns with project phases, fostering the financial discipline crucial for timely delivery, while the schedule’s parallel construction activities demonstrate efficient resource use and realistic deadlines. Sustainability has been embedded throughout, with solar PV, water recycling, and native landscaping reducing environmental footprint and operational costs.

Together, these components confirm that the project is feasible, economically viable, socially responsible, and well-positioned for successful delivery within the proposed timeframe — serving as a strong exemplar of effective construction planning and management using CPM techniques.